UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	3D animation does not use modelling.	3D animation relies on models.
A2	TRUE	A bitmap image is made up of pixels.	Bitmap images use a pixel grid.
A3	TRUE	A graphics tablet is used to draw directly into a computer.	A tablet is a drawing input device.
A4	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A5	TRUE	Copyright protects original multimedia works.	Copyright protects creative expression.
A6	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A7	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A8	TRUE	Hertz measures sound frequency.	Hertz is cycles per second.
A9	TRUE	Streaming allows media to begin playing before the whole file is downloaded.	Streaming delivers media continuously.
A10	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A11	TRUE	Tweening creates intermediate frames between keyframes.	Tweening fills the gaps.
A12	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A13	TRUE	Keyframes define important positions in an animation.	Keyframes mark major changes.
A14	TRUE	A higher sampling rate generally produces a more accurate recording.	More samples capture more detail.
A15	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	E	Storyboard	visual sequence plan
B2	J	Browser	software that displays web pages
B3	A	Frame rate	frames displayed per second
B4	I	Chroma key	replacing a colour background
B5	H	Virtual reality	simulated environment
B6	G	Streaming	playing media while it downloads
B7	M	Serif	font with small strokes at character ends
B8	F	Multimedia	integration of text, graphics, audio, video and animation
B9	L	Keyframe	important frame in an animation
B10	C	Bitmap	image made of pixels
B11	D	Linear multimedia	presentation without user control of sequence
B12	O	Card-based tool	authoring tool organised like pages or cards
B13	B	Vector	image made of mathematical objects
B14	N	Prototype	early model of a product
B15	K	RGB	additive colour model used by screens

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	A	Intended users
C2	D	Modelling and rendering
C3	D	Assemble multimedia elements
C4	B	Links to other content
C5	C	Display web pages
C6	B	Loudness
C7	C	Scanner
C8	A	The user can control content and flow
C9	B	Temporary working memory
C10	D	WAV
C11	A	A visual sequence plan
C12	A	CMYK
C13	A	Space between individual character pairs
C14	A	Pixels
C15	C	Text, graphics, audio, video and animation
C16	A	One shape into another
C17	A	Drawing each frame by hand
C18	C	Card-based
C19	B	Frequency
C20	D	Plotter
C21	C	Compiler
C22	A	It combines visuals and sound
C23	A	Merging of media and computing
C24	A	Printed output
C25	A	A video compression family
C26	A	An early model of a product
C27	B	Multimedia with hyperlinks
C28	A	Play while downloading
C29	B	Musical notes and instructions
C30	D	Another resource

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PART D - WORKED ANSWERS

QUESTION 1

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(4 marks)

QUESTION 2

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(5 marks)

QUESTION 3

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

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QUESTION 4

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

END OF MARKING SCHEME

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